Algebra 1 — Name____________________

Unit 11 — Date____________________

Lesson 4 Practice — Period____________________

Use the scenario below for questions 1-5.

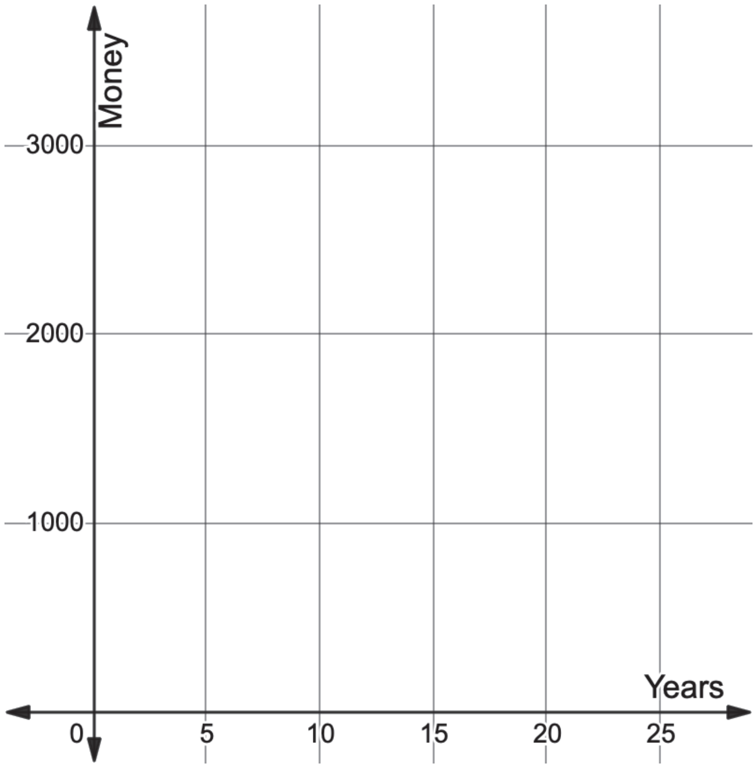
Janice started a bank account for her daughter with $\$625$, which earns 4.75% interest per year. The amount in her daughters account after x years can be modeled by the function $V(x) = \$625(1.0475)^x$.

1. Does the equation $V(x) = \$625(1.0475)^x$ represent exponential growth or decay?

2. What is the y-intercept and what does it represent in this situation?

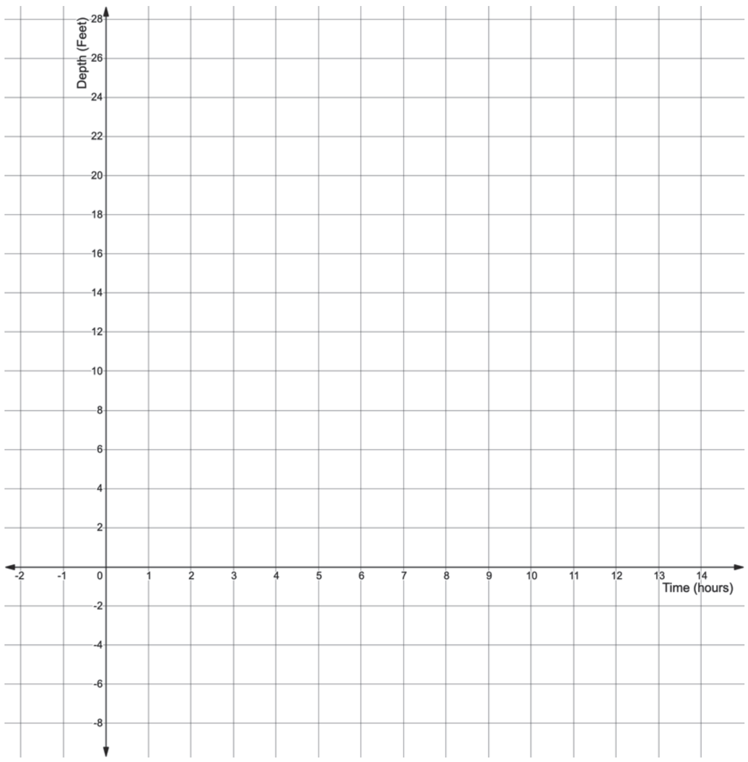
3. Explain why the value of b is not 4.75, the interest rate.

4. Graph the equation $V(x) = \$625(1.0475)^x$.



The equation $y = 14\left(\frac{4}{5}\right)^x$ represents the depth of the water as a pool drains in feet in x hours. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6. Domain		
7. Range		
8. y-intercept		
9. Constant Percent Rate of Change		
10. End Behavior		